

# COE 1195: Advanced Digital Design

## *Lab 4 – 32-bits MIPS CPU*

## *Milestone 5*

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# Milestone 5

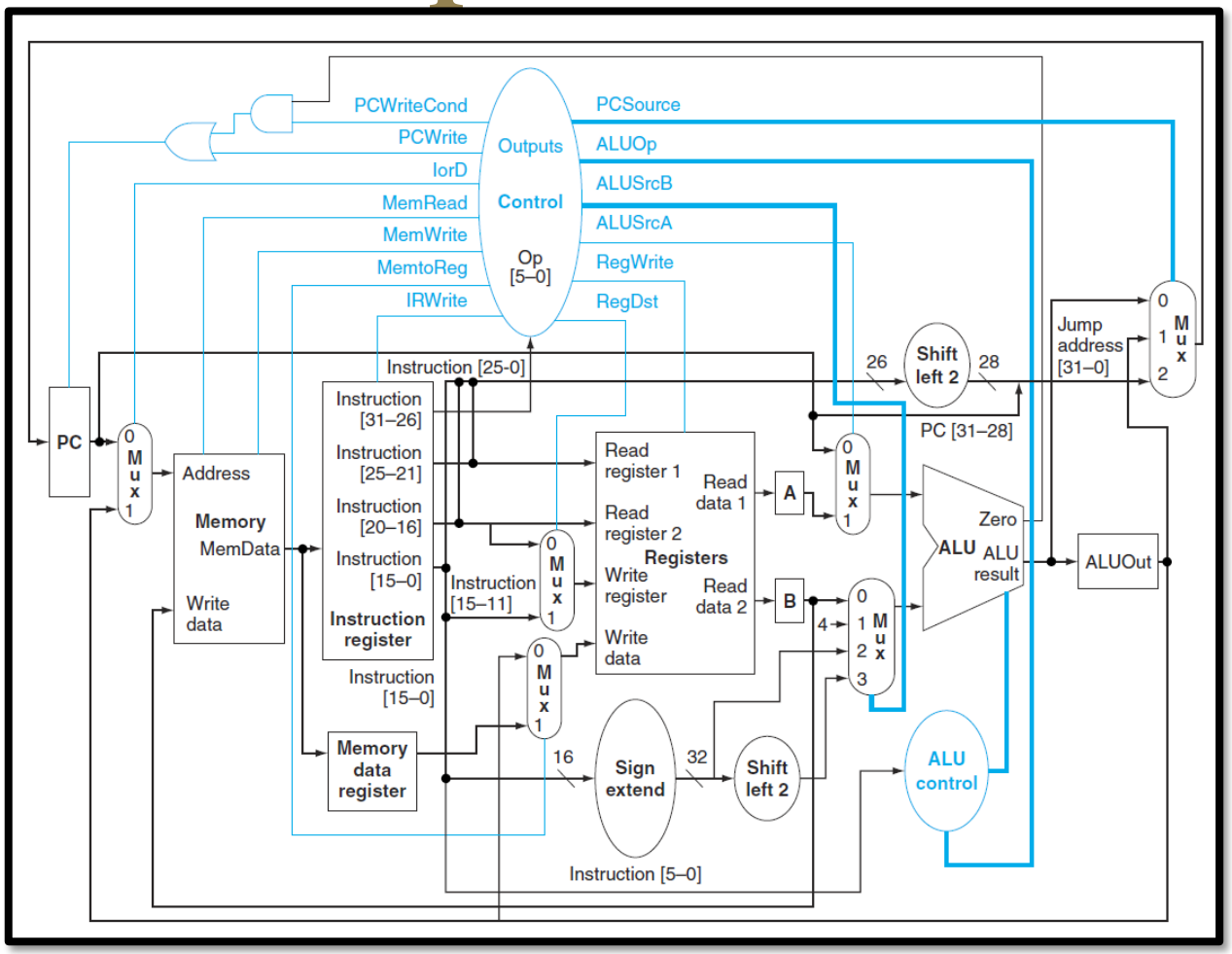
1. Implement Multiplication operations

2. Integrating with the provided memory

# Multiplication Instructions

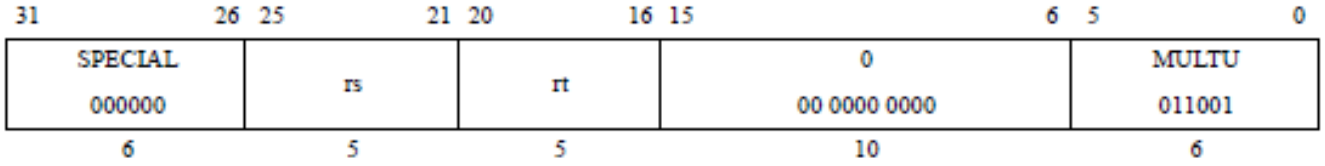
| | | |
|---|---|---|
| ADDI | ADDU | AND |
| ORI | SUB | SRA |
| SLLV | SLL | SLTI |
| LUI | SW | LW |
| LH | LB | CLO |
| BNE | BLTZAL | J |
| JR | MULTU | MFHI |
| MFLO | | |

# Milestone 5 – Implement Multiplication operations




*MIPS Multicycle CPU Datapath*

# MULTU




**Format:** `MULTU rs, rt` MIPS32 (MIPS I)

**Purpose:**

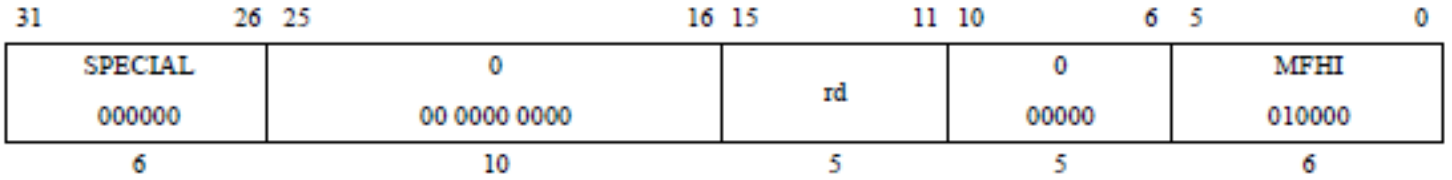
To multiply 32-bit unsigned integers

**Description:** `(LO, HI) ← rs × rt`

The 32-bit word value in GPR *rt* is multiplied by the 32-bit value in GPR *rs*, treating both operands as unsigned values, to produce a 64-bit result. The low-order 32-bit word of the result is placed into special register *LO*, and the high-order 32-bit word is placed into special register *HI*.

No arithmetic exception occurs under any circumstances.

# MFHI




Format: `MFHI rd` MIPS32 (MIPS I)

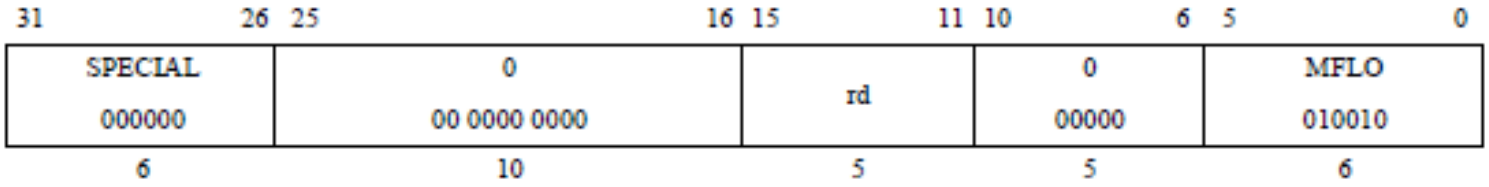
**Purpose:**

To copy the special purpose *HI* register to a GPR.

**Description:** `rd ← HI`

The contents of special register *HI* are loaded into GPR *rd*.

# MFLO




Format: `MFLO rd` MIPS32 (MIPS I)

**Purpose:**

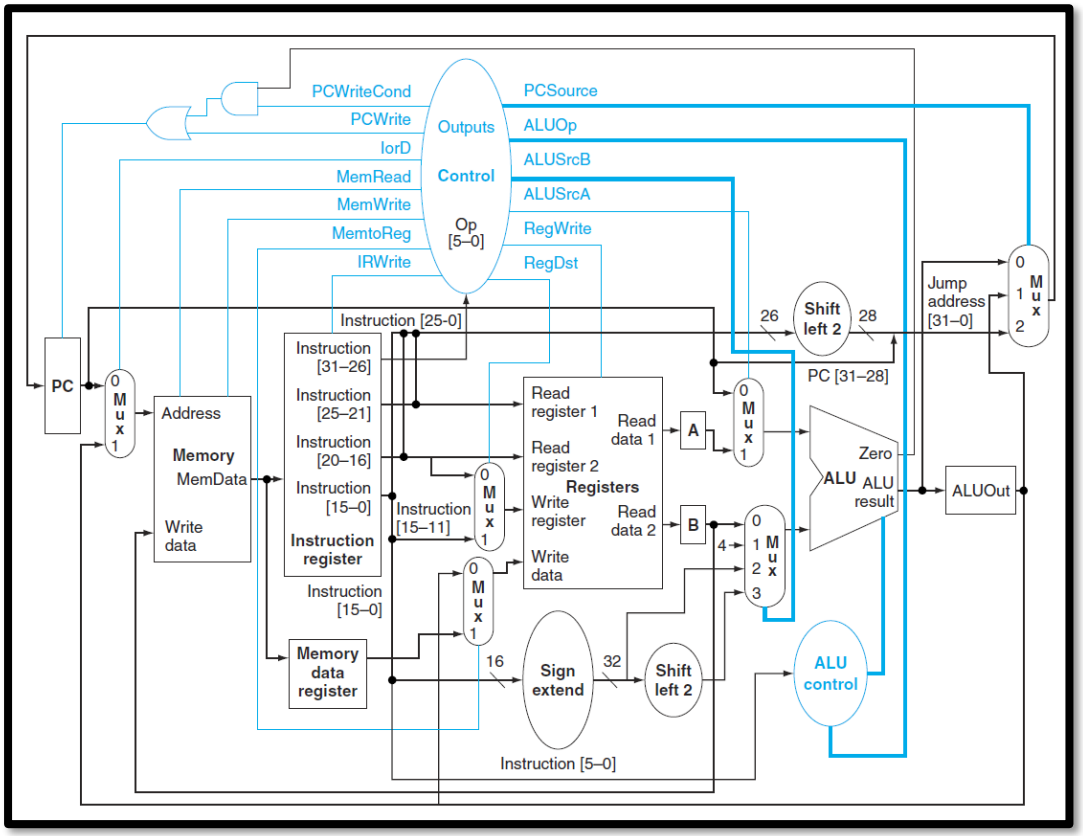
To copy the special purpose *LO* register to a GPR.

**Description:** `rd ← LO`

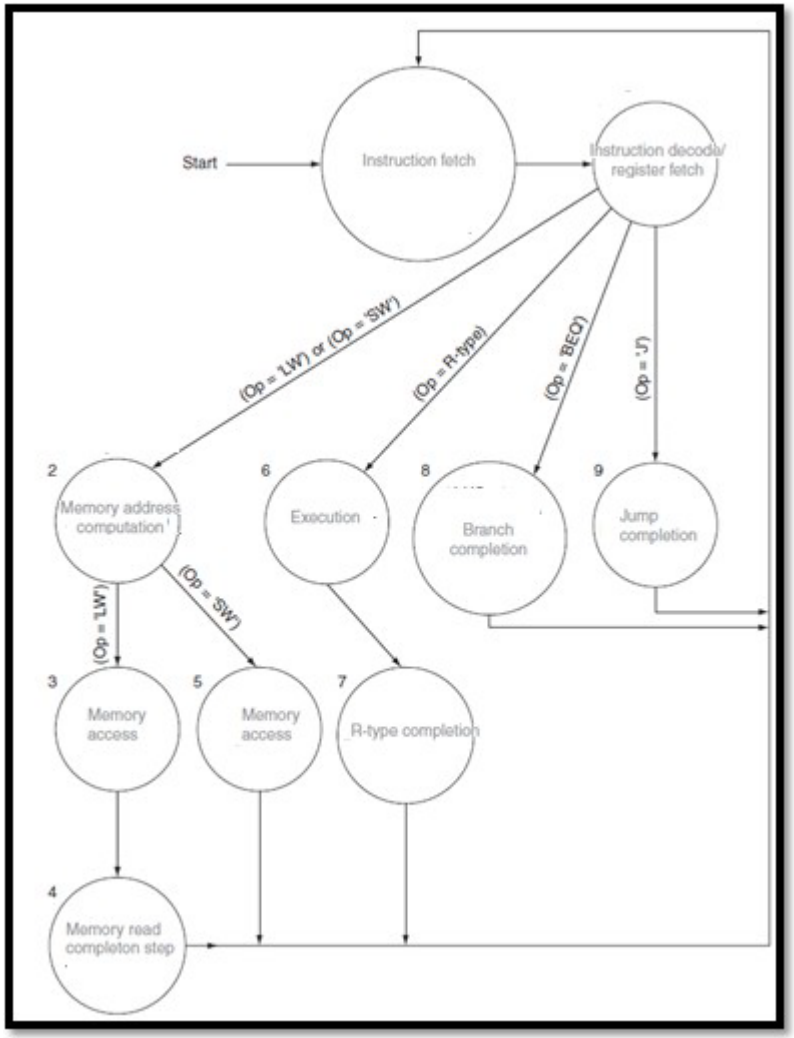
The contents of special register *LO* are loaded into GPR *rd*.

# Datapath and Control

## *MIPS Multicycle CPU FSM*




*MIPS Multicycle CPU Datapath*




*MIPS Multicycle CPU FSM Controller*